

Modeling the Determinants of Systolic Blood Pressure Using Multiple Linear Regression in R

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Abstract

This project investigates factors associated with systolic blood pressure using multiple linear regression in R. The analysis uses the Framingham Heart Study data and focuses on how demographic, behavioral, and clinical variables relate to blood pressure. Exploratory analysis suggested that systolic blood pressure was right-skewed, motivating transformation of the response. After fitting an initial full model, diagnostic checks indicated non-normality and heteroscedasticity, so a log transformation of systolic blood pressure was adopted. Backward elimination was then applied to obtain a final model. The final results show that age, diastolic blood pressure, prevalent hypertension, blood pressure medication use, glucose, cholesterol, heart rate, education, sex, and ten-year coronary heart disease risk are important predictors.

Keywords: multiple linear regression, Framingham Heart Study, systolic blood pressure, log transformation, diagnostics, model selection, R

1 Problem and Motivation

High blood pressure is one of the most important indicators of cardiovascular risk. Elevated systolic blood pressure is associated with heart disease, stroke, kidney problems, and other long-term health complications. Because blood pressure reflects the

combined influence of demographic, behavioral, and clinical characteristics, it is a natural outcome for a multiple linear regression study. In this project, we use data from the Framingham Heart Study to examine how factors such as age, sex, blood pressure medication use, cholesterol, glucose, diastolic blood pressure, and ten-year coronary heart disease risk are associated with systolic blood pressure.

This problem is important because it connects statistical modeling with a meaningful public health question. Rather than focusing only on whether a predictor is “significant,” the goal is to understand which factors are most strongly related to higher systolic blood pressure and whether those relationships remain after accounting for other variables. The project also provides an opportunity to demonstrate the full regression workflow emphasized in STAT 432: exploratory analysis, model fitting, assumption checking, response transformation, variable selection, and interpretation. By building a final model with reasonable diagnostics and clear interpretation, we can provide a practical summary of how major health indicators are related to blood pressure in this sample.

2 Data Description

The data used in this project come from the Framingham Heart Study. After removing observations with missing values, the working data set contained 3,658 observations and 16 variables. The response variable of interest is systolic blood pressure, denoted by `sysBP`, measured in mmHg. Candidate predictors considered in the analysis were:

- `male`: indicator for sex,
- `age`: age in years,
- `education`: education category,
- `currentSmoker`: indicator for current smoking,
- `cigsPerDay`: number of cigarettes smoked per day,
- `BPMeds`: indicator for blood pressure medication use,
- `prevalentStroke`: indicator for prior stroke,
- `prevalentHyp`: indicator for prevalent hypertension,
- `diabetes`: indicator for diabetes,
- `totChol`: total cholesterol,
- `diaBP`: diastolic blood pressure,
- `BMI`: body mass index,
- `heartRate`: heart rate,
- `glucose`: glucose level,
- `TenYearCHD`: indicator for ten-year coronary heart disease risk.

Summary statistics indicated that systolic blood pressure ranged from 83.5 to 295 mmHg, with a mean of 132.4 and median of 128.0. The histogram of systolic blood pressure showed a clear right-skewed distribution, with most observations concentrated between 110 and 140 mmHg and a long upper tail. Scatterplots suggested a positive association between age and systolic blood pressure, and a weaker positive association

between BMI and systolic blood pressure. A boxplot comparing smokers and non-smokers showed substantial overlap between the two groups, suggesting that smoking status alone does not strongly separate blood pressure levels in this sample.

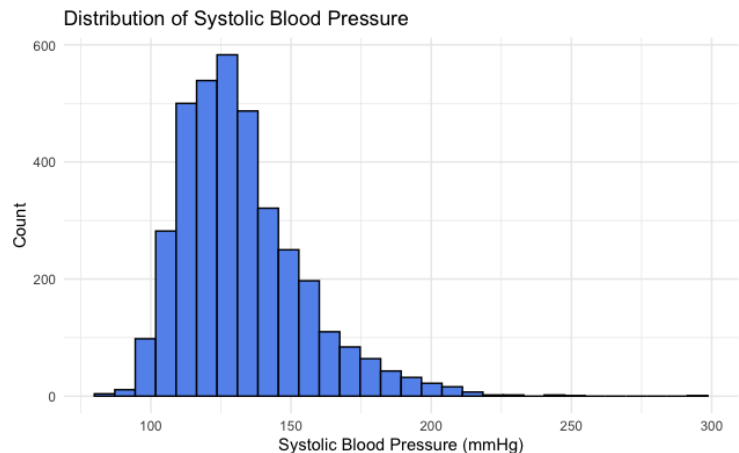


Fig. 1 Distribution of systolic blood pressure (sysBP) in the dataset. The histogram shows a right-skewed distribution, with most observations between 110–140 mmHg and a long tail toward higher values. The distribution is right-skewed, indicating the presence of higher-value outliers and suggesting that a transformation of the response variable may be appropriate for regression analysis.

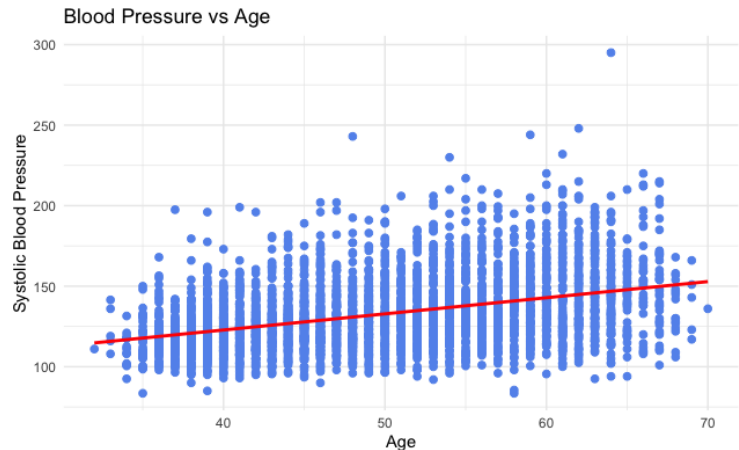


Fig. 2 Scatterplot of systolic blood pressure (sysBP) versus age. The plot shows a positive relationship, indicating that blood pressure tends to increase as age increases.

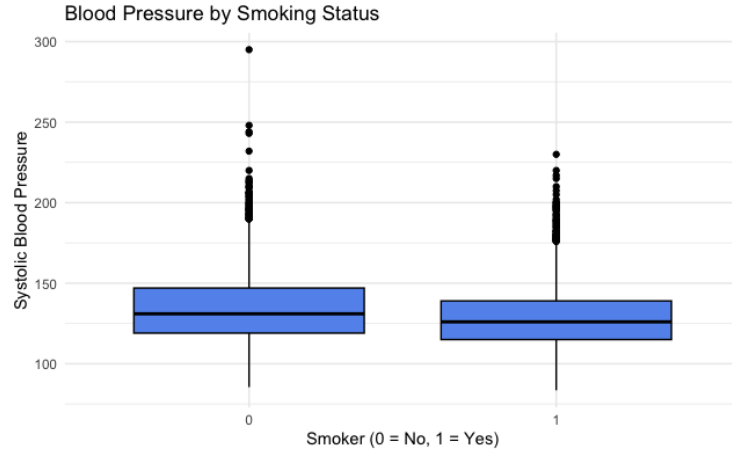


Fig. 3 Comparison of systolic blood pressure (sysBP) between smokers and non-smokers. The box-plot shows similar median values and overlapping distributions, suggesting no strong difference in blood pressure based on smoking status alone.

3 Questions of Interest

This project addresses the following scientific questions:

1. Which demographic and health-related factors are most strongly associated with higher systolic blood pressure in adults?
2. After accounting for other relevant health factors, how are age, hypertension status, diastolic blood pressure, and related clinical measurements associated with systolic blood pressure?

These questions are stated in scientific language rather than statistical language, consistent with the project guidelines. The analysis is designed to identify which factors are most important for explaining blood pressure variation and to determine whether those relationships persist after adjusting for the other variables in the model.

4 Regression Analysis, Results and Interpretation

4.1 Initial Multiple Linear Regression Model

Prior to model fitting, the dataset was partitioned into a training set and a testing set to allow for model validation. Specifically, 70% of the data were randomly assigned to the training set and the remaining 30% to the testing set using the `caret` package in R. The `createDataPartition` function was used to preserve the distribution of systolic blood pressure across both subsets.

Let Y_i denote systolic blood pressure for subject i , and let x_{ij} denote the j th predictor for subject i . The initial multiple linear regression model can be written as

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \quad (1)$$

where ε_i are random errors with mean zero and constant variance. Using the full set of candidate predictors in the training data, the initial model was fit in R as

$$\begin{aligned} \text{sysBP} \sim & \text{male} + \text{age} + \text{education} + \text{currentSmoker} \\ & + \text{cigsPerDay} + \text{BPMeds} + \text{prevalentStroke} + \text{prevalentHyp} \\ & + \text{diabetes} + \text{totChol} + \text{diaBP} + \text{BMI} \\ & + \text{heartRate} + \text{glucose} + \text{TenYearCHD}. \end{aligned} \quad (2)$$

The full model explained a large portion of the variation in systolic blood pressure; however, diagnostic checks indicated that key regression assumptions were not fully satisfied on the original scale of the response variable.

4.2 Exploratory Analysis and Need for Transformation

Exploratory analysis was used to assess the shape of the response and the relationships between systolic blood pressure and the candidate predictors. The right-skewed histogram of systolic blood pressure and the preliminary diagnostic plots suggested that a transformation of the response variable might be useful. In particular, the residuals from the original full model showed noticeable deviation from normality in the upper tail and mild heteroscedasticity.

To support the transformation choice, a Box–Cox style power transformation was examined for the response. The estimated transformation parameter was approximately $\lambda = -0.364$, with confidence bounds that did not support leaving the response untransformed. Although the suggested power was closer to -0.5 than to 0 , a log transformation was chosen because it provided a simpler and more interpretable scale while substantially improving the model diagnostics.

4.3 Model on the Transformed Response

The transformed model was fit using

$$\log(\text{sysBP}_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i. \quad (3)$$

After transformation, the residual plots improved considerably. The residuals versus fitted plot showed less curvature and more stable spread, and the normal Q–Q plot showed substantial improvement in the central region. Formal tests still detected slight departures from ideal assumptions, but the violations were mild enough to proceed.

Normality.

The Shapiro–Wilk test for the final transformed model produced $W = 0.99128$ with $p = 2.457 \times 10^{-11}$. This indicates that the residuals are not perfectly normal; however, the Q–Q plot showed only a mild upper-tail deviation, so the practical violation of normality was minor.

Equal variance.

The non-constant variance score test produced $\chi^2 = 15.78729$ with $p = 7.0877 \times 10^{-5}$, indicating evidence of heteroscedasticity. Even so, the residuals versus fitted plot suggested that the remaining departure from constant variance was mild and substantially improved relative to the untransformed model.

Independence and linearity.

Because the observations represent different individuals rather than repeated measurements over time, independence was regarded as reasonable. The residuals versus fitted plot for the final model showed no strong systematic trend, so linearity was considered satisfied.

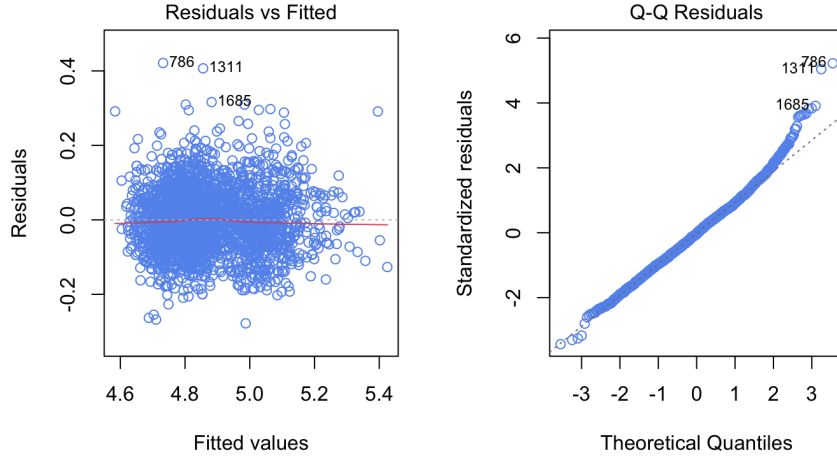


Fig. 4 Diagnostic plots for the final multiple linear regression model. The residuals vs. fitted plot shows no strong patterns, indicating that the linearity assumption is satisfied, while the Q-Q plot shows approximate normality with slight deviations in the upper tail.

4.4 Multicollinearity and Model Selection

Variance inflation factors were used to assess multicollinearity. For the final model, all VIF values were close to 1 and well below the common threshold of 5, indicating no evidence of serious multicollinearity. Backward elimination was then carried out on the transformed model using a removal criterion of 0.10 to avoid dropping potentially useful predictors too early. The final reduced model retained predictors that were all significant at the 0.05 level.

The final fitted model was

$$\begin{aligned}
\log(\widehat{\text{sysBP}}) = & 4.030 - 0.0177(\text{male}) + 0.002853(\text{age}) - 0.006130(\text{education}) \\
& + 0.04044(\text{BPMeds}) + 0.09207(\text{prevalentHyp}) + 0.00008838(\text{totChol}) \\
& + 0.007422(\text{diaBP}) + 0.0003727(\text{heartRate}) + 0.0003035(\text{glucose}) \\
& + 0.01730(\text{TenYearCHD}).
\end{aligned} \tag{4}$$

Equivalently, the model specification can be written as

$$\begin{aligned}
\log(\text{sysBP}) = & \beta_0 + \beta_1(\text{male}) + \beta_2(\text{age}) + \beta_3(\text{education}) \\
& + \beta_4(\text{BPMeds}) + \beta_5(\text{prevalentHyp}) + \beta_6(\text{totChol}) \\
& + \beta_7(\text{diaBP}) + \beta_8(\text{heartRate}) + \beta_9(\text{glucose}) \\
& + \beta_{10}(\text{TenYearCHD}) + \varepsilon.
\end{aligned} \tag{5}$$

The final model had $R^2 = 0.7378$ and adjusted $R^2 = 0.7368$, meaning that approximately 73.7% of the variation in log systolic blood pressure was explained by the retained predictors.

4.5 Interpretation of the Final Model

Because the response is on the log scale, each coefficient is interpreted approximately as a percentage change in systolic blood pressure, holding the other predictors fixed.

- **Age:** Each additional year of age is associated with about a 0.29% increase in systolic blood pressure, holding the other variables constant.
- **Male:** Being male is associated with about a 1.77% lower systolic blood pressure than being female, after adjustment for the other predictors.
- **Education:** Higher education level is associated with a slight decrease in systolic blood pressure, about 0.61% per education-category increase.
- **BPMeds:** Individuals taking blood pressure medication have about 4.0% higher systolic blood pressure on average, likely reflecting underlying cardiovascular risk.
- **Prevalent hypertension:** Subjects with prevalent hypertension have about 9.2% higher systolic blood pressure, making this one of the strongest predictors in the model.
- **Total cholesterol:** A one-unit increase in total cholesterol is associated with a very small increase in systolic blood pressure.
- **Diastolic blood pressure:** Diastolic blood pressure is the strongest continuous predictor in the final model, with each one-unit increase associated with about a 0.74% increase in systolic blood pressure.
- **Heart rate and glucose:** Both variables have positive but relatively small associations with systolic blood pressure.
- **Ten-year CHD risk:** Individuals classified as at risk for coronary heart disease have about 1.7% higher systolic blood pressure.

These results indicate that systolic blood pressure is most strongly related to age, diastolic blood pressure, and hypertension status, while cholesterol, heart rate, glucose, and long-term cardiovascular risk contribute smaller but still meaningful effects.

4.6 Leverage and Influential Observations

Leverage values and standardized residuals were examined to assess the influence of individual observations. Most observations had low leverage and standardized residuals between -2 and 2 . A small number of high-leverage points and outliers were present, but there were no observations that simultaneously showed both very high leverage and large residuals in a way that suggested serious influence on the model fit. Thus, no strong influential points were identified.

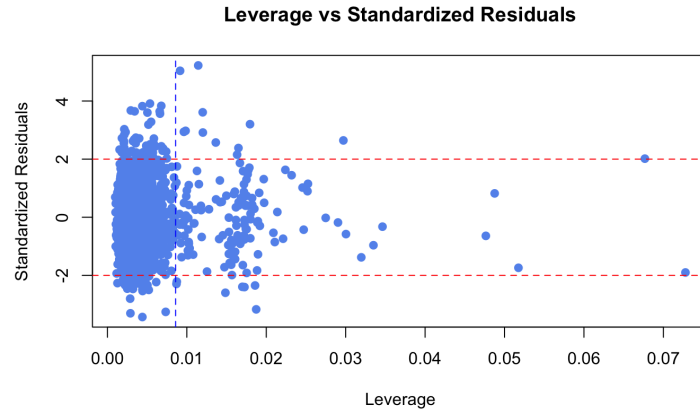


Fig. 5 Leverage versus standardized residuals for the final regression model. The majority of observations have low leverage and lie within the ± 2 standardized residual range. Although a few high leverage points and outliers are present, none appear to be simultaneously high leverage and large residual, indicating no influential observations.

5 Conclusions

This study developed a multiple linear regression model for systolic blood pressure using demographic and clinical variables from the Framingham Heart Study. The final model showed that age, diastolic blood pressure, and prevalent hypertension were the strongest predictors of systolic blood pressure, while medication use, glucose, cholesterol, heart rate, sex, education, and ten-year CHD risk also contributed to explaining blood pressure variation. The final adjusted R^2 of 0.7368 indicates that the model explained a substantial portion of the variation in the transformed response.

A key part of the analysis was the response transformation. The original model showed noticeable departures from normality and constant variance, but the log transformation improved both diagnostics substantially. Although formal tests still detected mild violations, the graphical diagnostics indicated that the final model assumptions

were reasonably satisfied. The final model was also free of multicollinearity and showed no severe influential observations.

Overall, the primary message is that systolic blood pressure is closely connected to age and cardiovascular health indicators, especially diastolic blood pressure and hypertension status. These findings are reliable for the current sample, though generalizability is limited to populations similar to the Framingham study participants. Future work could compare alternative response transformations, explore interaction terms, or assess predictive performance on external data.

Supplementary information. Supplementary material may include the full R code, additional exploratory plots, preliminary diagnostic plots for models that were not retained, and detailed hypothesis test output.

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Declarations

Funding

No external funding was reported for this study.

Conflict of interest/Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

The Framingham Heart Study data analyzed in this project were provided for course use. Derived outputs and figures may be shared by the authors upon request.

Materials availability

Not applicable.

Code availability

The R code used for cleaning, exploratory analysis, model fitting, diagnostics, transformation, and variable selection is included in the appendices.

Author contribution

William Lorenzo contributed to the statistical analysis, model interpretation, and manuscript drafting. Drake Hashimoto contributed to project development, interpretation, and review. Anthony Barnes contributed to project development, interpretation, and review.

Appendix A R Code

The following R code was used for data preprocessing, exploratory data analysis, model fitting, transformation, diagnostics, model selection, and validation.

Package Loading

```
# Loading Packages
library(tidyverse)
library(car)
library(caret)
library(GGally)
```

Data Import and Cleaning

```
df <- read_csv("framingham.csv")
df <- na.omit(df)
```

Exploratory Data Analysis

```
summary(df)
names(df)

ggplot(df, aes(x = sysBP)) +
  geom_histogram(bins = 30, col = "black", fill = "cornflowerblue") +
  labs(title = "Distribution of Systolic Blood Pressure")
```

Data Splitting

```
set.seed(452)
train_index <- createDataPartition(df$sysBP, p = 0.7, list = FALSE)
train_set <- df[train_index, ]
test_set <- df[-train_index, ]
```

Initial Model

```
mlr1 <- lm(sysBP ~ ., data = train_set)
summary(mlr1)
```

Diagnostic Checks (Initial Model)

```
par(mfrow = c(1,2))
plot(mlr1, 1:2)

shapiro.test(mlr1$residuals)
ncvTest(mlr1)
```

Response Transformation

```
summary(powerTransform(mlr1))

mlr1_log <- lm(log(sysBP) ~ ., data = train_set)
summary(mlr1_log)
```

Diagnostics of Transformed Model

```
plot(mlr1_log, 1:2)

shapiro.test(mlr1_log$residuals)
ncvTest(mlr1_log)
```

Multicollinearity Check

```
vif(mlr1_log)
```

Backward Elimination

```
mlr2 <- update(mlr1_log, . ~ . - diabetes)
mlr3 <- update(mlr2, . ~ . - BMI)
mlr4 <- update(mlr3, . ~ . - prevalentStroke)
mlr5 <- update(mlr4, . ~ . - currentSmoker)
mlr_final <- update(mlr5, . ~ . - cigsPerDay)

summary(mlr_final)
```

Final Model Diagnostics

```
plot(mlr_final, 1:2)

shapiro.test(mlr_final$residuals)
ncvTest(mlr_final)
vif(mlr_final)
```

Leverage and Influence Analysis

```
h <- hatvalues(mlr_final)
r <- rstandard(mlr_final)

p <- length(coef(mlr_final)) - 1
n <- nrow(train_set)

cutoff <- 2*(p + 1)/n

plot(h, r)
abline(h = c(-2, 2), col = "red", lty = 2)
abline(v = cutoff, col = "blue", lty = 2)

high_lev <- which(h > cutoff)
outliers <- which(abs(r) > 2)
influential <- which(h > cutoff & abs(r) > 2)
```

Final Model Summary

```
summary(mlr_final)
```

Appendix B Additional Figures and Diagnostics

This appendix contains supplementary plots and diagnostic results that support the modeling decisions described in the main text.

Original-Scale Model Diagnostics

The initial model fit on the original `sysBP` scale showed violations of the normality and constant variance assumptions. In particular, the residuals versus fitted plot suggested slight heteroscedasticity, and the Q-Q plot showed substantial deviation in the upper tail.

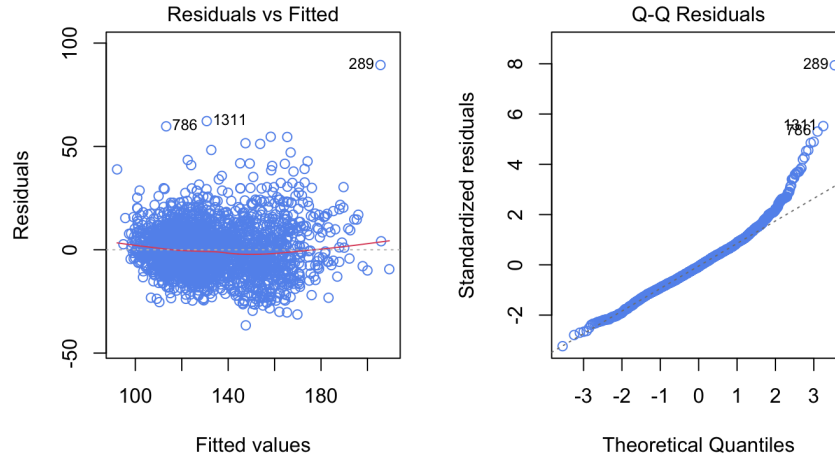


Fig. B1 Diagnostic plots for the initial multiple linear regression model fit on the original systolic blood pressure scale. The residuals vs. fitted plot suggests mild non-constant variance, and the Q-Q plot indicates non-normality in the upper tail.

The formal diagnostic tests also supported these concerns. The Shapiro–Wilk test rejected normality of residuals ($W = 0.96591$, $p < 2.2 \times 10^{-16}$), and the non-constant variance score test indicated heteroscedasticity ($\chi^2 = 294.8436$, $df = 1$, $p < 2.22 \times 10^{-16}$).

Transformation Evidence

To address the diagnostic violations, a Box–Cox power transformation was examined for the response variable. The estimated power was $\lambda = -0.3644$, with confidence

interval $(-0.5114, -0.2175)$, suggesting that a transformation of the response was appropriate. For interpretability and simplicity, a log transformation of systolic blood pressure was used.

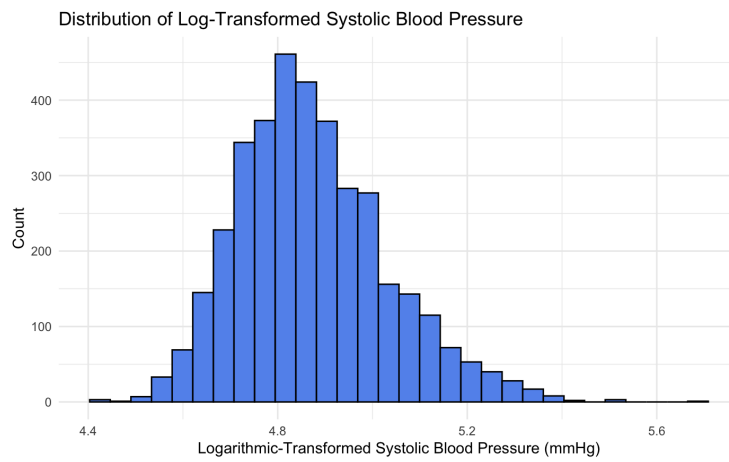


Fig. B2 Histogram of log-transformed systolic blood pressure. The log transformation reduces right skewness and produces a more symmetric distribution, supporting its use in the regression analysis.

Diagnostics After Response Transformation

After fitting the model with $\log(\text{sysBP})$ as the response, the diagnostics improved substantially. The residuals were more evenly scattered, and the Q-Q plot showed much closer agreement with the reference line, although slight deviations remained in the upper tail.

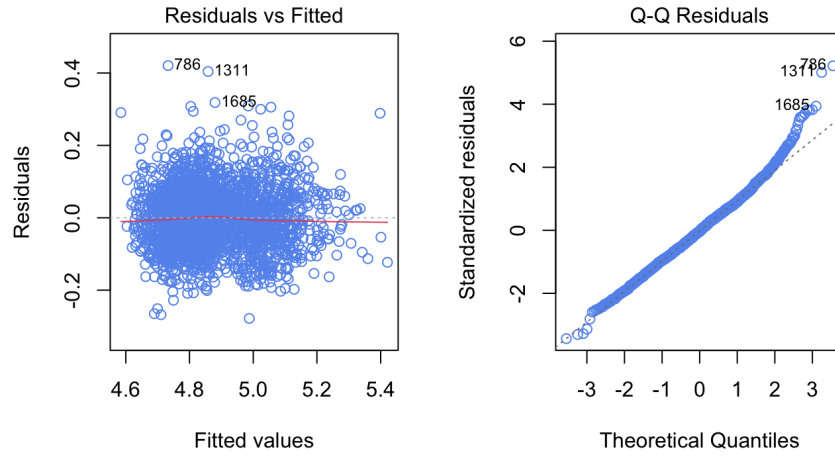


Fig. B3 Diagnostic plots for the transformed model using $\log(\text{sysBP})$ as the response. The transformation improved normality and variance stability relative to the original model.

The Shapiro–Wilk test for the transformed model remained statistically significant ($W = 0.99127$, $p = 2.429 \times 10^{-11}$), but the test statistic moved much closer to 1, indicating a substantial improvement in normality. Similarly, the non-constant variance score test remained significant ($\chi^2 = 16.73495$, $df = 1$, $p = 4.2982 \times 10^{-5}$), but the test statistic was dramatically reduced compared with the original model.

Exploratory Plots Not Central to the Main Text

Additional exploratory plots supported the relationships discussed in the main analysis. Age showed a positive association with systolic blood pressure, BMI showed a weaker positive trend, and smoking status showed substantial overlap between groups.

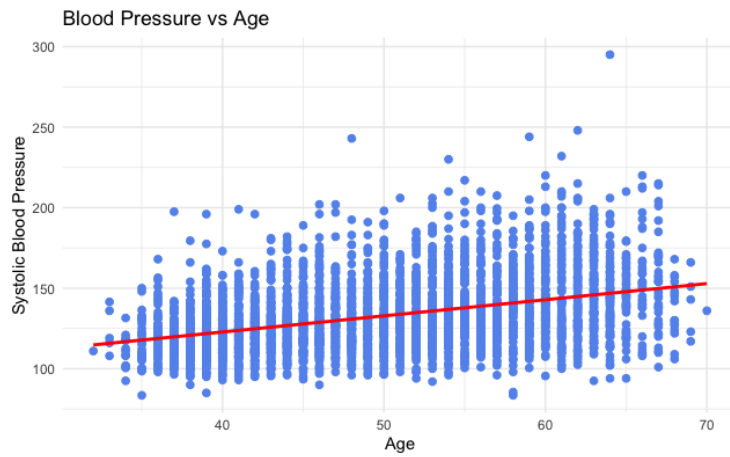


Fig. B4 Scatterplot of systolic blood pressure versus age. The fitted line indicates a positive association between age and systolic blood pressure.

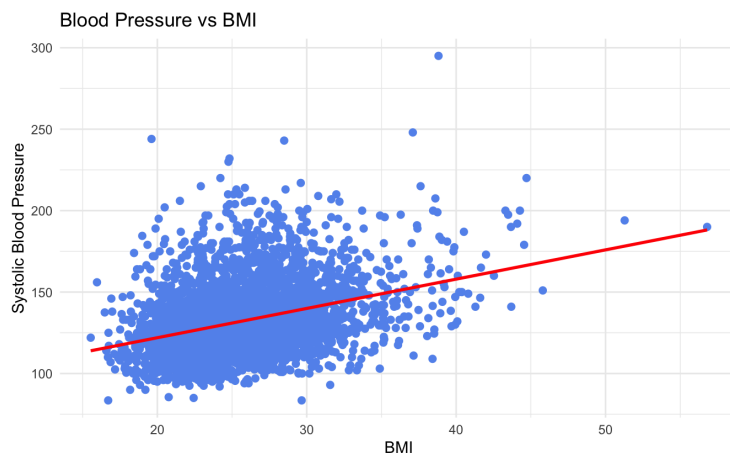


Fig. B5 Scatterplot of systolic blood pressure versus BMI. The fitted line suggests a mild positive association.

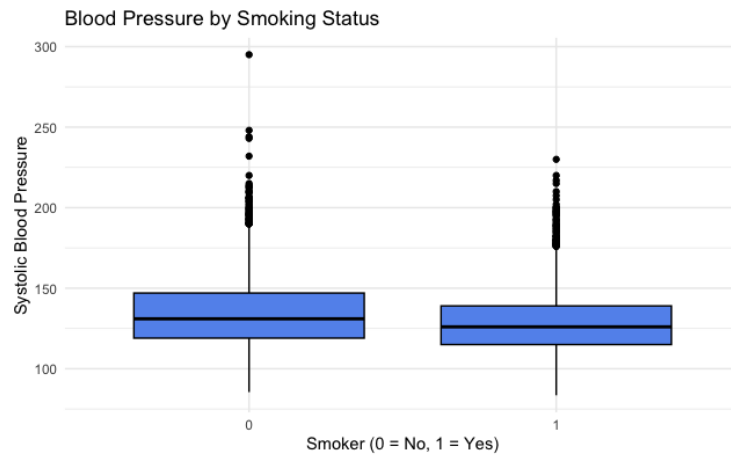


Fig. B6 Boxplot of systolic blood pressure by smoking status. The two groups show substantial overlap, suggesting no strong marginal difference based on smoking status alone.

Model Selection Support

Variance inflation factors for the transformed full model were all below 3, indicating no serious multicollinearity before model selection. Backward elimination was then performed using a significance level of 0.10. Variables removed sequentially were `diabetes`, `BMI`, `prevalentStroke`, `currentSmoker`, and `cigsPerDay`. The resulting final model retained only predictors that were significant at the 0.05 level.

Final Model Diagnostics and Influence Check

The final model showed reasonably satisfactory assumptions. Diagnostic plots indicated that linearity and independence were acceptable, while normality and equal variance showed only mild remaining departures.

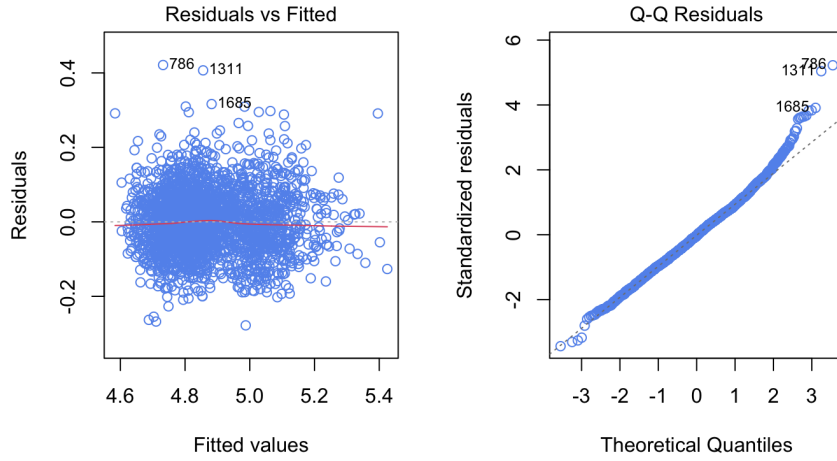


Fig. B7 Diagnostic plots for the final model. The residuals vs. fitted plot shows no strong systematic pattern, and the Q-Q plot indicates approximate normality with slight deviation in the upper tail.

The Shapiro–Wilk test for the final model gave $W = 0.99128$ with $p = 2.457 \times 10^{-11}$, indicating slight but mild non-normality. The non-constant variance score test gave $\chi^2 = 15.78729$, $df = 1$, and $p = 7.0877 \times 10^{-5}$, indicating mild heteroscedasticity. These violations were not severe and were much smaller than in the original model.

VIF values for the final model were all close to 1 and well below 5, indicating no evidence of multicollinearity.

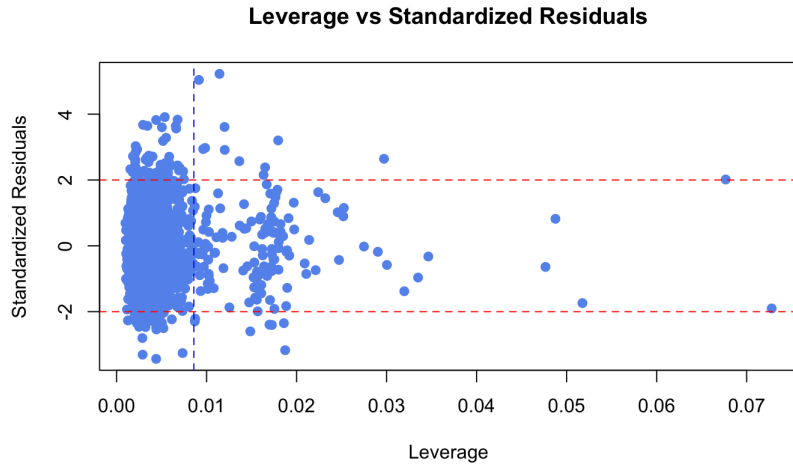


Fig. B8 Leverage versus standardized residuals for the final model. Most observations have low leverage and lie within the ± 2 residual range. Although a few high-leverage points and outliers are present, no strongly influential observations were identified.

The leverage cutoff was computed as $2(p+1)/n = 0.008587041$. While some observations exceeded the leverage threshold and others exceeded the ± 2 residual threshold, very few points satisfied both criteria simultaneously. Therefore, the final model was not unduly influenced by individual observations.

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